Project Design Phase-I
Solution Architecture

Date	08 March 2026
Team ID	
Project Name	Ucab – Cab Booking System
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture bridges business needs and technical implementation. For UCab, the goals are:

• Select the best technologies to solve real-time ride matching.
• Describe system structure, behavior, and operational qualities.
• Define features, development phases, and solution requirements.
• Provide specifications for defining, managing, and delivering the solution.

Architecture Overview

- Client: React SPA (Vite) with Leaflet maps and Socket.IO client
- API: Express REST with JWT authentication
- Database: MongoDB (users, drivers, bookings, payments, support)
- Realtime: Socket.IO namespaces/rooms per vehicle type and rider

Component Responsibilities

- Auth Service: registration, login, JWT issuance, middleware guards
- Booking Service: create booking, query bookings, status lifecycle
- Driver Service: availability, my rides, profile
- Payment Service: record payments and list history
- Support Service: ticket creation and updates

Data Flow (Request/Response & Events)

1. Rider posts booking → server validates and stores → server emits newRideRequest to vehicle room
2. Driver accepts → server updates booking → emits rideAccepted to rider room and rideRequestCancelled to other drivers
3. Ride progresses → status updates → completion and rating
4. Payment recorded → history available to both parties

Quality Attributes

- Performance: low-latency sockets; indexed queries
- Security: hashed passwords, token auth, least-privilege roles
- Scalability: horizontal scaling with sticky sessions or Redis adapter for sockets
- Observability: structured logs and error tracking (planned)

(Diagram placeholder) Insert architecture and data-flow diagram for UCab here.